

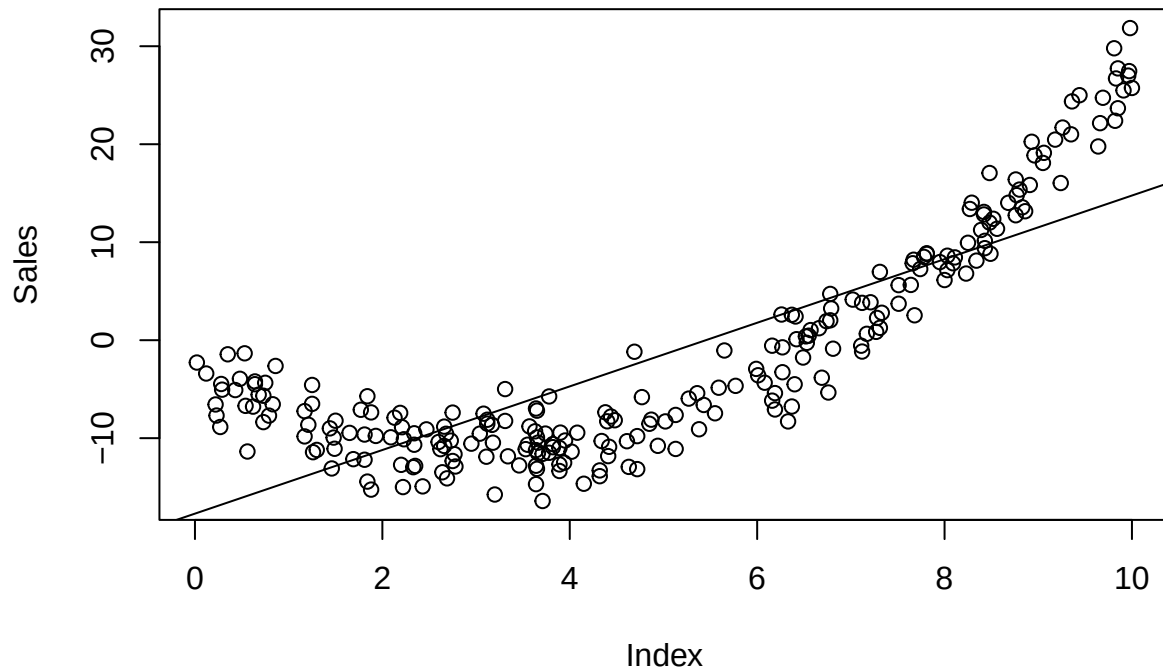
STAT2170 ASSIGNMENT PART II

MINH CHAU VU (48040509)

2024-05-17

Question 1 [25 marks] a. [3 marks] Load the data from the file sales.csv and create a scatter plot of Sales against Index. Comment on the relationship between the two variables.

```
sales_dat <- read.csv("sales.csv", header = TRUE)
plot(Sales ~ Index, data = sales_dat)
abline(lm(Sales ~ Index, data = sales_dat))
```

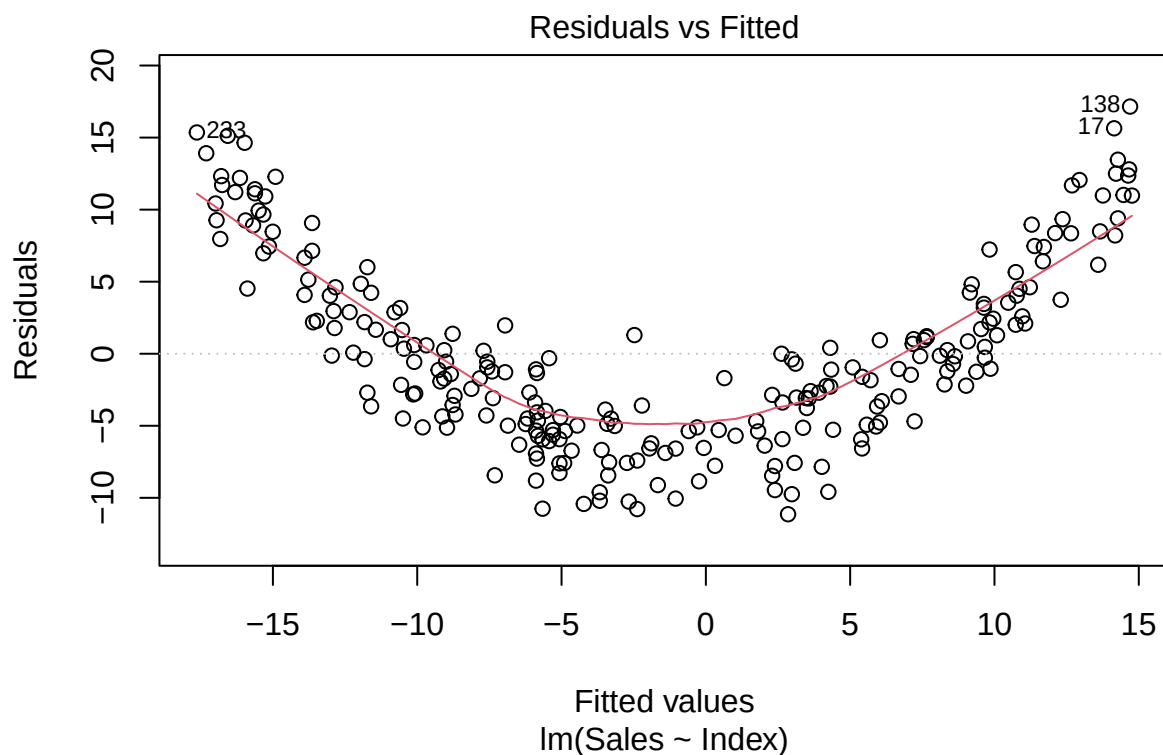


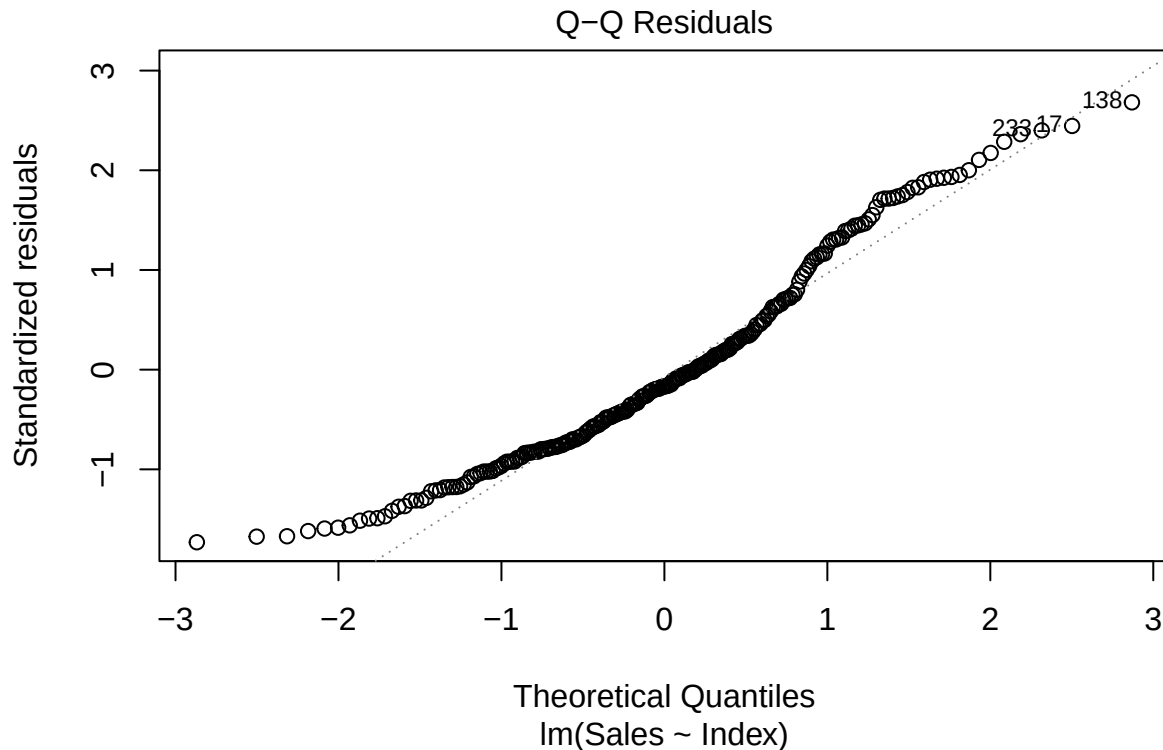
Comment: The scatter plot shows a positive relationship between Consumer Confidence Index and Change in retail Sales. It is clear that the graph is not likely linear since it does not follow a straight line, so quadratic or cubic relationship seems to be more suitable. b. [5 marks] Fit a simple linear regression model and named it as M1 to predict Sales using Index. Validate the model through diagnostic checks and comment.

```
M1 <- lm(Sales ~ Index, data = sales_dat)
summary(M1)
```

```
##
## Call:
## lm(formula = Sales ~ Index, data = sales_dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.139  -4.988  -1.086   4.028  17.152
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -17.7007     0.8278  -21.38  <2e-16 ***
## Index         3.2464     0.1444   22.48  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.45 on 241 degrees of freedom
## Multiple R-squared:  0.6772, Adjusted R-squared:  0.6759
## F-statistic: 505.6 on 1 and 241 DF,  p-value: < 2.2e-16
```

```
plot(M1, which = 1:2)
```





Comment: The model appear to be significant with P-Value < 0.05 and and R-Squared value of 0.6759. The normal QQ plot show a nice linear trend, so the normality assumption seems to be valid. However, there is curvation and a clear pattern appear in the residual plot, which means the linearity assumption does not hold.

```
M2 <- lm(Sales ~ Index + I(Index^2), data = sales_dat)
summary(M2)
```

```
##
## Call:
## lm(formula = Sales ~ Index + I(Index^2), data = sales_dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.755 -1.967  0.037  1.749  7.827
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.50608    0.50308  -6.969 3.06e-11 ***
## Index       -4.96591    0.23046 -21.548 < 2e-16 ***
## I(Index^2)   0.80875    0.02201  36.744 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.511 on 240 degrees of freedom
## Multiple R-squared:  0.9513, Adjusted R-squared:  0.9509
## F-statistic: 2343 on 2 and 240 DF, p-value: < 2.2e-16
```

```
M3 <- lm(Sales ~ Index + I(Index^2) + I(Index^3), data = sales_dat)
summary(M3)
```

```
##
## Call:
## lm(formula = Sales ~ Index + I(Index^2) + I(Index^3), data = sales_dat)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
##	-5.7850	-1.9384	0.0545	1.7424	7.8321

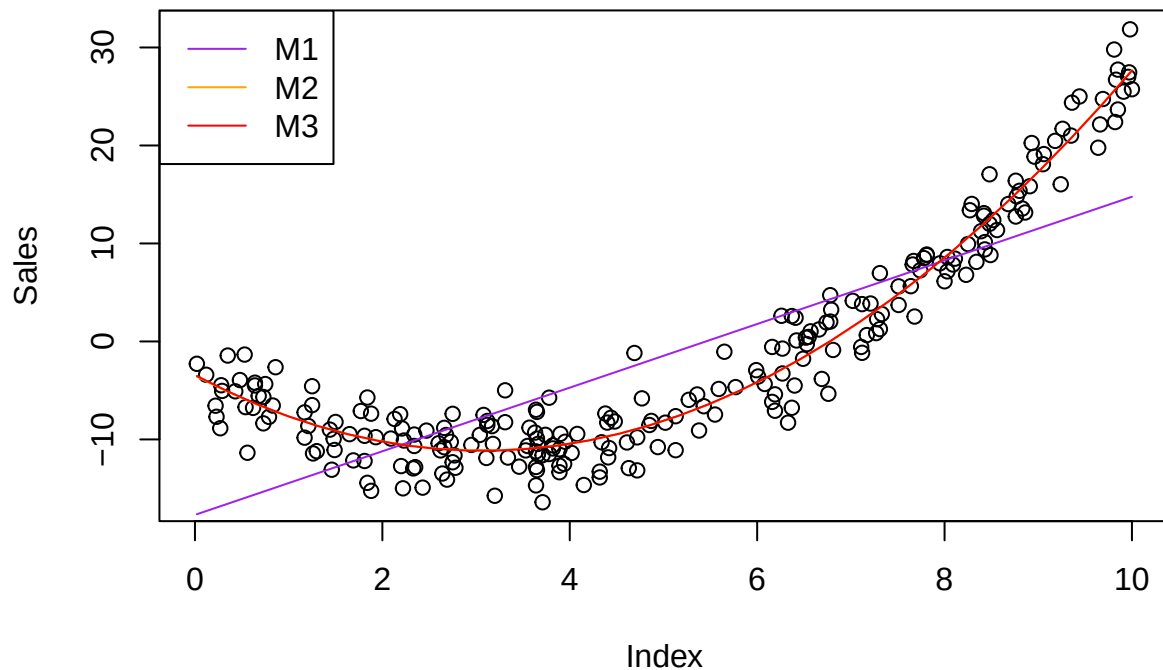
```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-3.421148	0.668122	-5.121	6.27e-07 ***
## Index	-5.062632	0.550206	-9.201	< 2e-16 ***
## I(Index^2)	0.832770	0.125982	6.610	2.48e-10 ***
## I(Index^3)	-0.001599	0.008255	-0.194	0.847

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.516 on 239 degrees of freedom
## Multiple R-squared:  0.9513, Adjusted R-squared:  0.9507
## F-statistic: 1556 on 3 and 239 DF,  p-value: < 2.2e-16
```

Comment: It is clear that M2 and M3 models show improvements compared to M1 with R-squared value approximately 0.95, partly indicating that they fit the data much better than the linear model. However, in model 3, the cubic term provide insignificant result (P-Value = 0.847>0.05). Whereas, model 2, seems to be highly significant. d. [5 marks] Plot the data and add the three predicted lines from models M1, M2, and M3 to your plot. Comment on the fit of the models.

```
x = seq(from = min(sales_dat$Index), to = max(sales_dat$Index), length = 32)
dataoflines = data.frame(Index = x)
yhat = predict(M2, newdata = dataoflines)
plot(Sales ~ Index, data = sales_dat)
lines(x, yhat, col = "orange")
yhat.single = predict(M1, newdata = dataoflines)
lines(x, yhat.single, col = "purple")
yhat.cubic = predict(M3, newdata = dataoflines)
lines(x, yhat.cubic, col = "red")
legend("topleft", legend = c("M1", "M2", "M3"), col = c("purple", "orange", "red"), lty = 1)
```



Comment: The linear model (M1) captures the upward trend but cannot capture the curvature of the data. Meanwhile, M2 and M3 models fit the data better. M2 and M3 models seem to follow the same pattern. This could be explained by the approximately similar R-Squared value of both models. e. [4 marks] Assess the significance of the linear, quadratic and cubic terms in M3 using a Sequential Sum of Squares (Hint: A sequential ANOVA is necessary.) Comment on the results.

```
anova(M3)
```

```
## Analysis of Variance Table
##
## Response: Sales
##           Df Sum Sq Mean Sq  F value Pr(>F)
## Index      1 21036.0  21036.0 3322.5228 <2e-16 ***
## I(Index^2)  1   8513.8   8513.8 1344.7051 <2e-16 ***
## I(Index^3)  1     0.2     0.2    0.0375 0.8466
## Residuals 239   1513.2     6.3
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

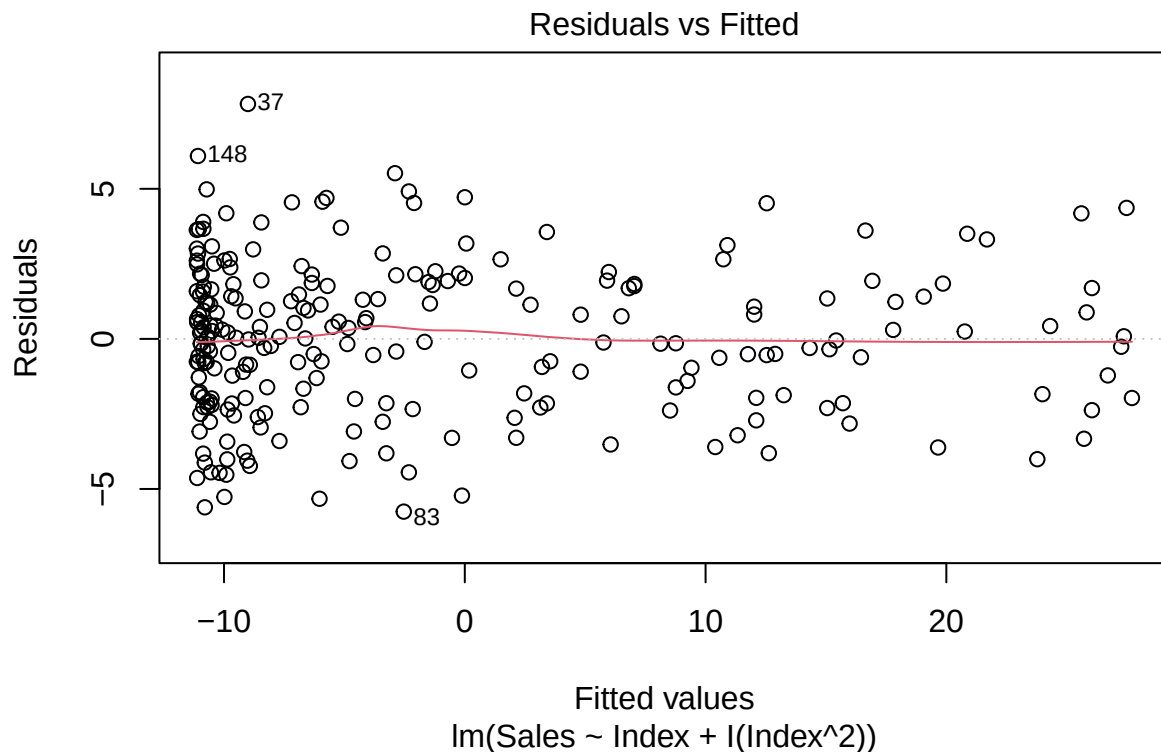
Comment: The linear and quadratic terms in the model appear significant with P-Value $< 2e-16$ (< 0.05). Meanwhile, cubic term is insignificant (P-Value = 0.8466 > 0.05). Therefore, it is necessary to refit the model without the cubic term. f. [4 marks] Choose the best model among M1, M2, and M3 and validate it. Provide reasoning and comment on the model fit. - The best model among M1,M2,M3 is M2

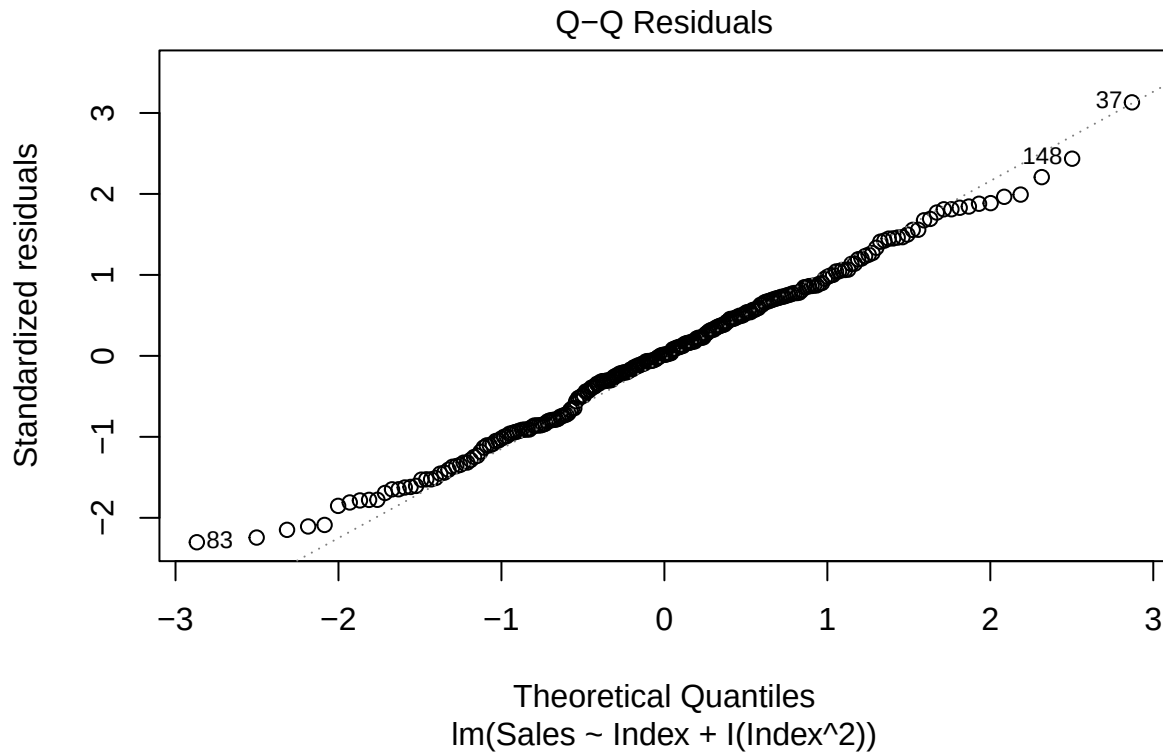
```
anova(M2)
```

```
## Analysis of Variance Table
##
## Response: Sales
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Index       1 21036.0  21036.0  3335.9 < 2.2e-16 ***
## I(Index^2)   1   8513.8   8513.8  1350.1 < 2.2e-16 ***
## Residuals  240   1513.4     6.3
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Reasoning: +) M1 model is not validated with residual plot having curvation. R-Squared value is lower compared with M2, M3 +) M3 model includes insignificant cubic term (P-Value = 0.8466 < 0.05), therefore, cubic term needs to be removed +) M2 has a high R-Squared value (=0.9513) and Adjusted R-Squared Value (=0.9509) => M2 is the best model among M1,M2 and M3
- Validate the model:

```
plot(M2, which = 1:2)
```





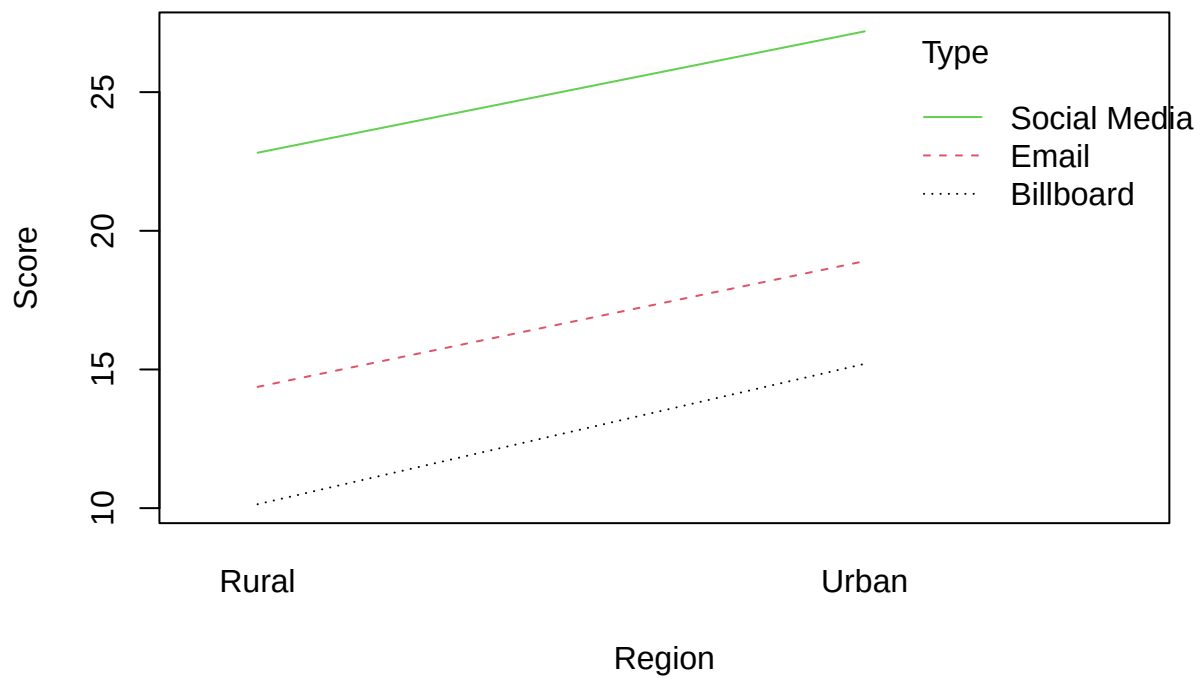
Comment: The quantile plot of residuals close to linear, which means the errors are close to normally distributed. The residuals vs fitted plot does not show any apparent pattern, suggesting that the quadratic fit is valid and sufficient.

Question 2 [30 marks] a. [5 marks] Construct two different preliminary graphs that investigate different features of the data and comment. Loading the data:

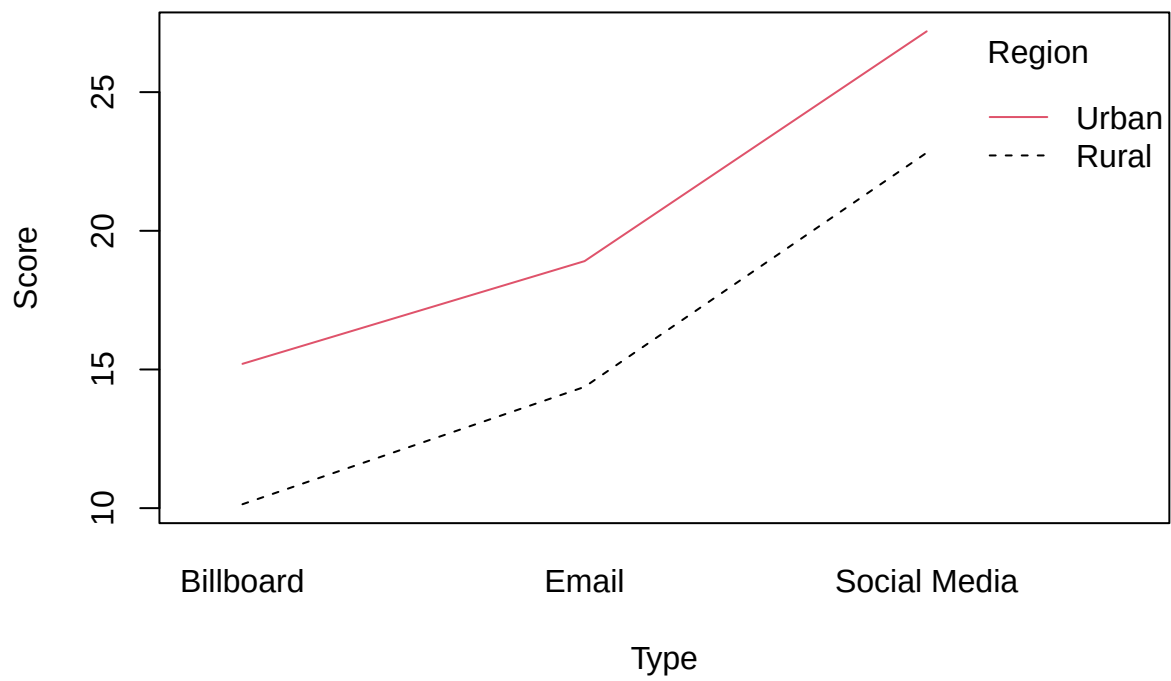
```
campaign_dat <- read.csv ("campaign.csv", header = TRUE)
```

Construct two preliminary graphs

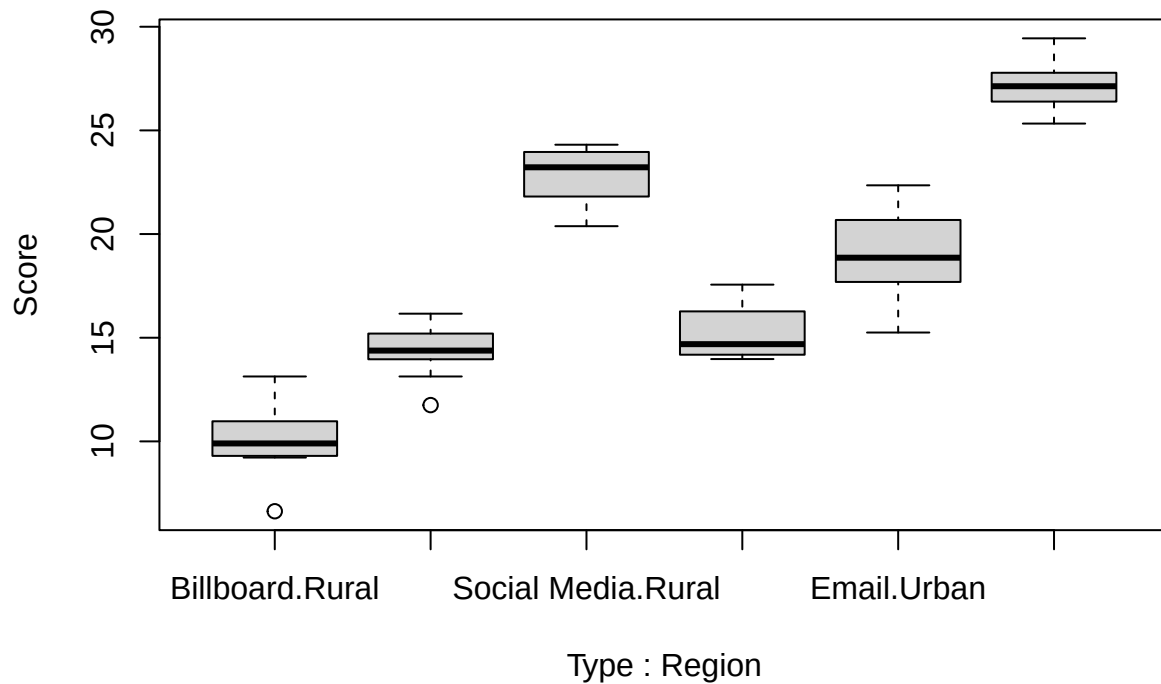
```
interaction.plot(campaign_dat$Region,campaign_dat$Type, campaign_dat$Score, trace.label = "Type", xlab = "Region", ylab = "Score")
```



```
interaction.plot(campaign_dat$Type, campaign_dat$Region, campaign_dat$Score, trace.label = "Region", xlab = "Region", ylab = "Score", main = "Interaction Plot of Campaign Type and Region")
```

```
boxplot(Score ~ Type + Region, data = campaign_dat)
```



Comment: For interaction plots, we can observe parallel lines for each group at various levels of independent variables. This suggests that there is no significant interaction effect between Type and Region on Score. For boxplot, the equal variance assumption seems to be valid due to similar boxplot size.

- b. [6 marks] Write down the full interaction model for this situation, defining all appropriate parameters. The full interaction model: $Y_{ijk} = \mu + i + j + ij + ijk$, $ijk \sim N(0, 2)$ where the parameters are: Y_{ijk} : the **Score** response i : the **Region** effect, there are two levels: Urban and Rural j : the **Type** effect, there are three levels: Billboard, Email and Social Media ij : interaction effect between **Region** and **Type**. $ijk \sim N(0, 2)$ is the unexplained variation
- c. [8 marks] Analyse the data to study the effect of Type and Region on the percentage increase in engagement Score at 5% significance level. Remember to • state the null and alternative hypothesis for each test, • check assumptions (model diagnostics), • interpret the results.
 - Hypothesis: $H_0 : ij = 0$ for all i, j against $H_1 : \text{at least one } ij \neq 0$

*Fitting this interaction model:

```
campaign.dat <- lm (Score ~ Region * Type, data = campaign_dat)
anova(campaign.dat)
```

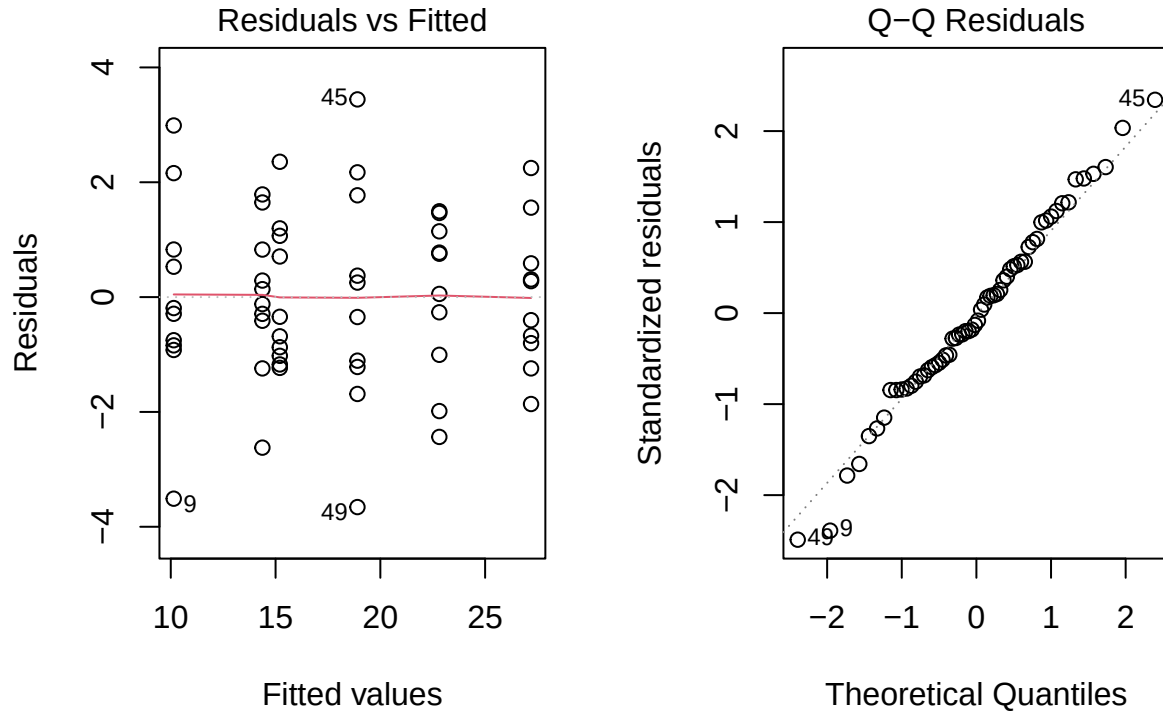
```
## Analysis of Variance Table
##
## Response: Score
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Region      1  325.45   325.45  135.7281 2.336e-16 ***
```

```
## Type          2 1585.09  792.54 330.5242 < 2.2e-16 ***
## Region:Type  2    1.29    0.64  0.2683    0.7657
## Residuals    54 129.48    2.40
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can see that the interaction term is not significant ($P\text{-value} = 0.7657 > 0.05$), therefore, we have not yet reach our final model and interaction can be removed.

- Try checking model diagnostics:

```
par(mfrow = c(1, 2))
plot(campaign.dat, which = 1:2)
```



Comment: The QQ plot follow a nice linear trend and the residual plot shows equal spread around fitted values. So, both normal assumption and and constant variance assumption are appropriate.

- d. [7 marks] Repeat the above test analysis for the main effects.

- Analysis for main effects:

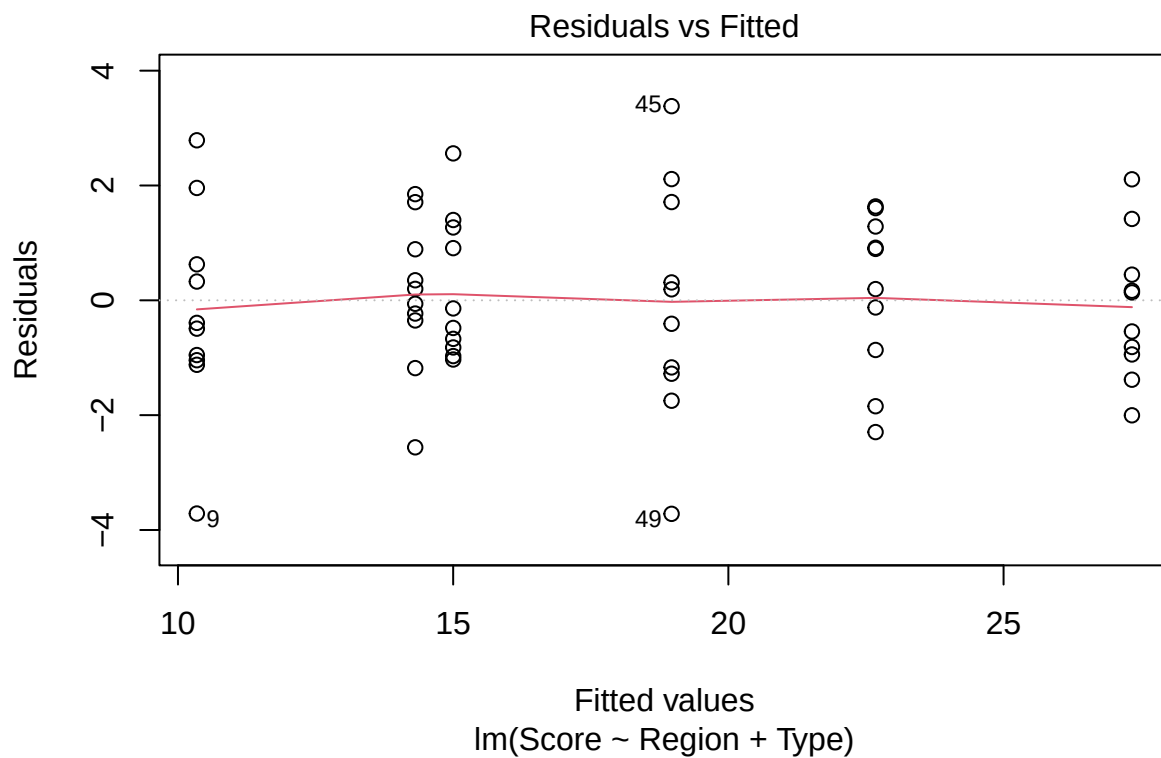
```
campaign.dat1 = lm (Score ~ Region + Type, data = campaign_dat)
campaign.dat1 = update(campaign.dat, . ~ . - Region:Type)
anova(campaign.dat1)
```

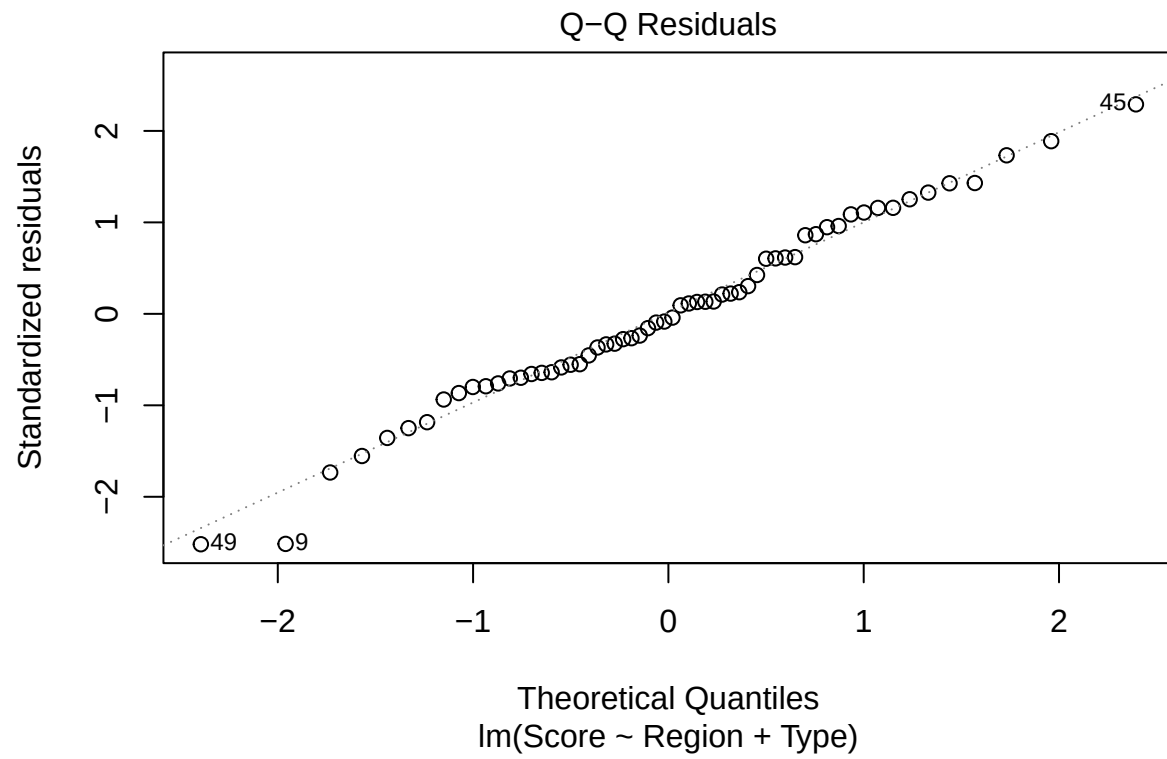
```
## Analysis of Variance Table
##
## Response: Score
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Region      1  325.45   325.45  139.37 < 2.2e-16 ***
## Type        2 1585.09   792.54  339.39 < 2.2e-16 ***
## Residuals   56  130.77     2.34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comment: Both of the main effects are significant ($P\text{-value} = < 2.2e-16 < 0.05$)

- Model diagnostic:

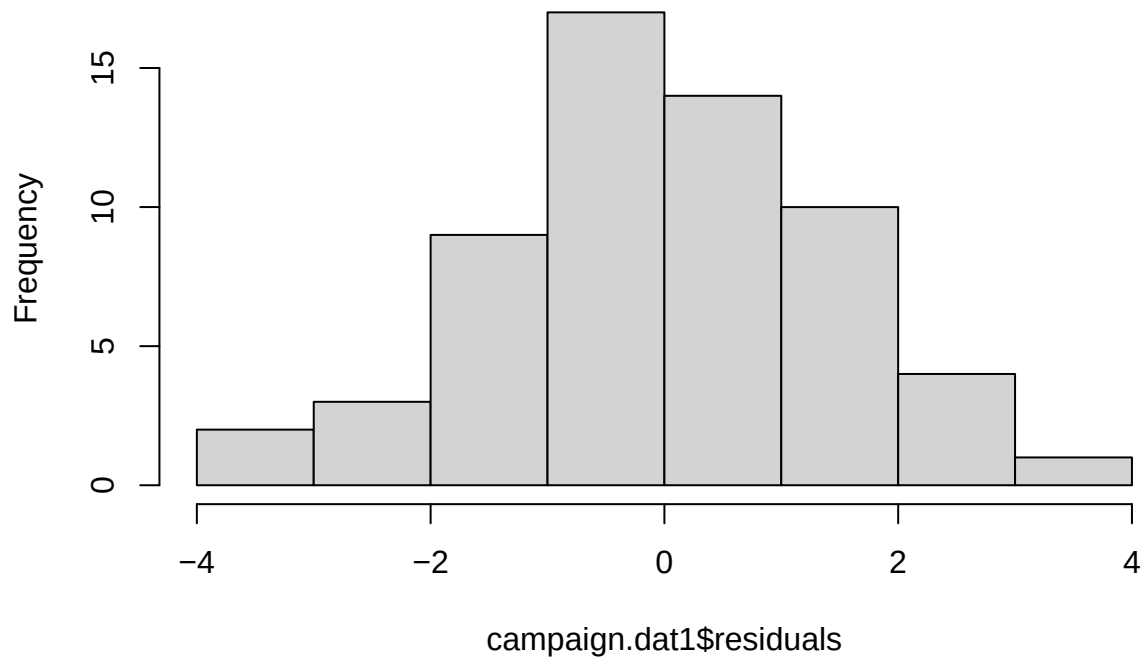
```
plot(campaign.dat1, which = 1:2)
```





```
hist(campaign.dat1$residuals)
```

Histogram of campaign.dat1\$residuals



Comment: There is no pattern in residuals, quantile plot of residuals shows linear trend and residuals are normally distributed. All of the assumptions are valid and the model is sufficient.

- e. [4 marks] Using TukeyHSD produce multiple comparisons between each level for both Type and Region. Comment on the effectiveness of the marketing campaign type on customer engagement scores and also the impact of region on customer engagement scores. (Hint: Confirm the design is balanced before proceeding with the TukeyHSD test.)

*Check if the test is balanced:

```
table(campaign_dat[c("Type", "Region")])
```

```
##           Region
## Type      Rural Urban
## Billboard      10   10
## Email          10   10
## Social Media   10   10
```

Comment: A design is balanced when all replicates are the same for each group. Therefore, this design is balanced with $r = 10$

- TukeyHSD on Score and Type:

```
clov.aov=aov(Score ~ Type, data = campaign_dat)
TukeyHSD(clov.aov)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = Score ~ Type, data = campaign_dat)
##
## $Type
##          diff          lwr          upr      p adj
## Email-Billboard      3.9675  1.814604  6.120396 0.0001247
## Social Media-Billboard 12.3315 10.178604 14.484396 0.0000000
## Social Media-Email      8.3640  6.211104 10.516896 0.0000000
```

Comment: - The comparison between Email and Billboard is significant ($P\text{-value} = 0.0001247 < 0.05$), indicating that Email campaigns lead to higher percentage increase in engagement Score compared to Billboard ones. - The comparison between Social Media and Billboard is highly significant ($P\text{-value} = 0 < 0.05$), indicating that Social Media campaigns lead to higher percentage increase in engagement Score compared to Billboard ones. - - The comparison between Social Media and Email is highly significant ($P\text{-value} = 0 < 0.05$), indicating that Social Media campaigns lead to higher percentage increase in engagement score compared to Email ones. => In conclusion, ranking the effectiveness of each type of campaign on the increase in engagement score, we get: Social Media > Email > Billboard

- - TukeyHSD on Score and Region:

```
clov.aov2 = aov(Score ~ Region, data = campaign_dat)
TukeyHSD(clov.aov2)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = Score ~ Region, data = campaign_dat)
##
## $Region
##          diff          lwr          upr      p adj
## Urban-Rural 4.658 1.84685 7.46915 0.0015755
```

Comment: The comparison between Urban and Rural regions is significant ($p \text{ adj} < 0.05$), indicating that Urban regions have a higher engagement score than Rural regions. Therefore, campaigns would be more effective when being held across Urban regions.